

TrES-5b

R-2748

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_260729 · created 2026-09-02T23:02:36+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2461250.7014 +/- 0.0046 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.185 +/- 0.053
Transit depth (Rp/Rs)^2	3.41 +/- 1.97 [%]
Orbital Inclination (inc)	85.04 +/- 2.85
Airmass coefficient 1 (a1)	2.46 +/- 1.34
Airmass coefficient 2 (a2)	-0.83 +/- 1.14
Scatter in the residuals of the lightcurve fit is	58.72 %
Best Comparison Star	None
Optimal Aperture	2.41
Optimal Annulus	7.25
Transit Duration (day)	0.076 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.185 ± 0.053 vs 0.142 (deviation 0.81σ)
Duration fit vs published	0.076 d vs 0.0758 d (deviation 0.01σ)
Mid-time O–C	8.7 min
Depth SNR	3.5
Residual scatter	58.72 %

Light curve

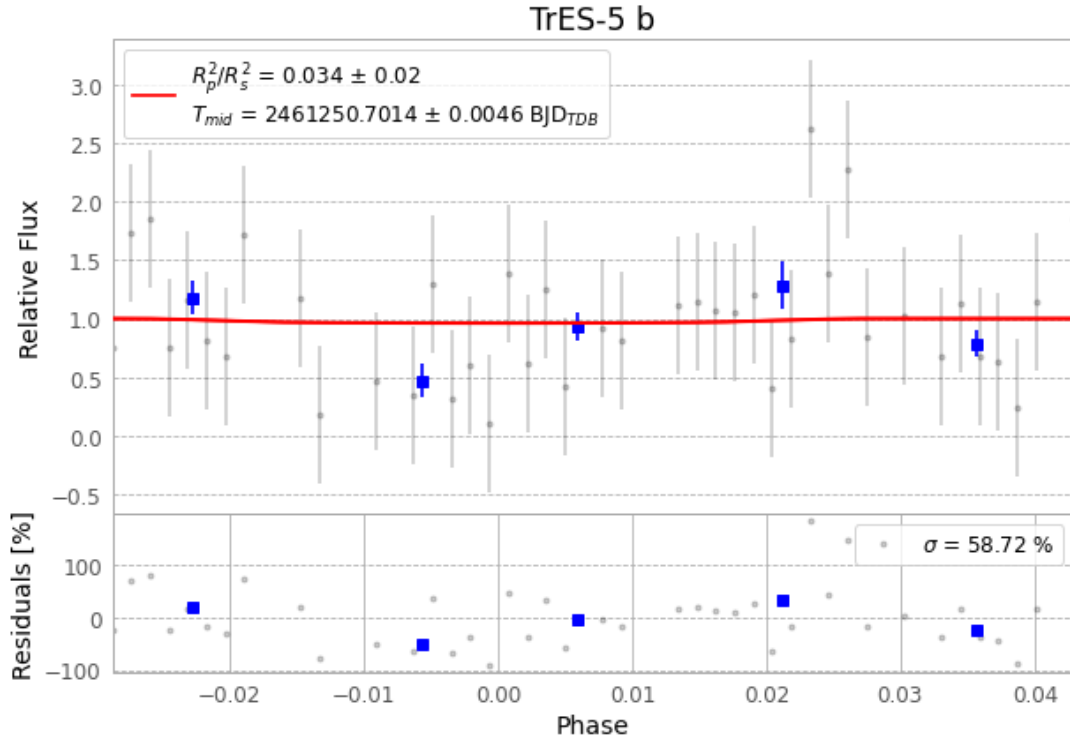


Figure 1 — FinalLightCurve_TrES-5 b_2026-07-28.png (SHA-256 1cc5233d4f9d0a2a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [442, 326], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a86c7179d2661d17...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

55 FITS frames (36 MB), TRES-5260729034519.FITS → TRES-5260729062716.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-260729.log (07c0e1e27b0c...), FinalLightCurve_TrES-5 b_2026-07-28.png (1cc5233d4f9d...), FinalLightCurve_TrES-5 b_2026-07-28.csv (d5be0f14e9b5...)

Compute resources

Wall time 186.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-02 23:02Z.